Intrusion Detection Systems (IDS) and Traffic Analysis on Linux

Lecture note for hands on practice by A. Akande.
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Install and Configure Snort

Install Snort:

Open your terminal on the VM where Snort will run and type:

```
sudo apt-get update
sudo apt-get install snort
```

During installation, it'll ask for the **network interface to monitor** — enter your active interface name (for example: `eth0` or `ens33` or `wlan0` — you can check your interfaces by running:

```
ip addr
```

Configure Snort

Open the Snort configuration file:

```
sudo nano /etc/snort/snort.conf
```

Find and Set `HOME_NET`

Use `CTRL + W` in nano to search for `HOME_NET`
Locate this line:

```
var HOME_NET any
```

And replace `any` with your OWASP VM IP address or your desired subnet.

Example:

```
var HOME_NET 192.168.56.101
```

Uncomment and Adjust Relevant Rules

Scroll down and make sure at least one relevant rule is uncommented (remove the `#` at the beginning of the line).
For example:

```
include $RULE_PATH/local.rules
```

Or enable specific attack rules under `community-rules` or `local.rules` if you've created custom rules.

Save with `CTRL + O`, press `ENTER`, then exit with `CTRL + X`.

Start Snort in IDS Mode

Run Snort in IDS Mode:

Replace `<interface>` with your actual interface (like `eth0`)

```
sudo snort -A console -c /etc/snort/snort.conf -i eth0
```

- `-A console` → output alerts to terminal
- `-c` → specifies config file
- `-i` → specifies interface to listen on

Snort will start and display messages for traffic it detects based on the rules.

Generate Network Traffic

Install hping3 (if not already installed)

```
sudo apt-get install hping3
```

Launch a SYN Flood Attack towards the OWASP VM

From your attacker VM (or the same VM, targeting the OWASP VM IP):

```
hping3 -S -p 80 192.168.56.101 --flood
```

Explanation:

- `-S` → set SYN flag
- `-p 80` → target port 80 (web)
- `--flood` → send packets as fast as possible

Note: You can omit `--flood` if you just want to test basic detection without overwhelming traffic.

Monitor Traffic with Wireshark

Open Wireshark

- Select the same network interface that Snort is monitoring (e.g., `eth0`)
- Click **Start Capture**

Set a TCP Filter

In the Wireshark filter bar, type:

```
tcp
```

Then press **Enter**

Now you should see TCP traffic — including your SYN flood packets.

You can also filter specifically for SYN packets:

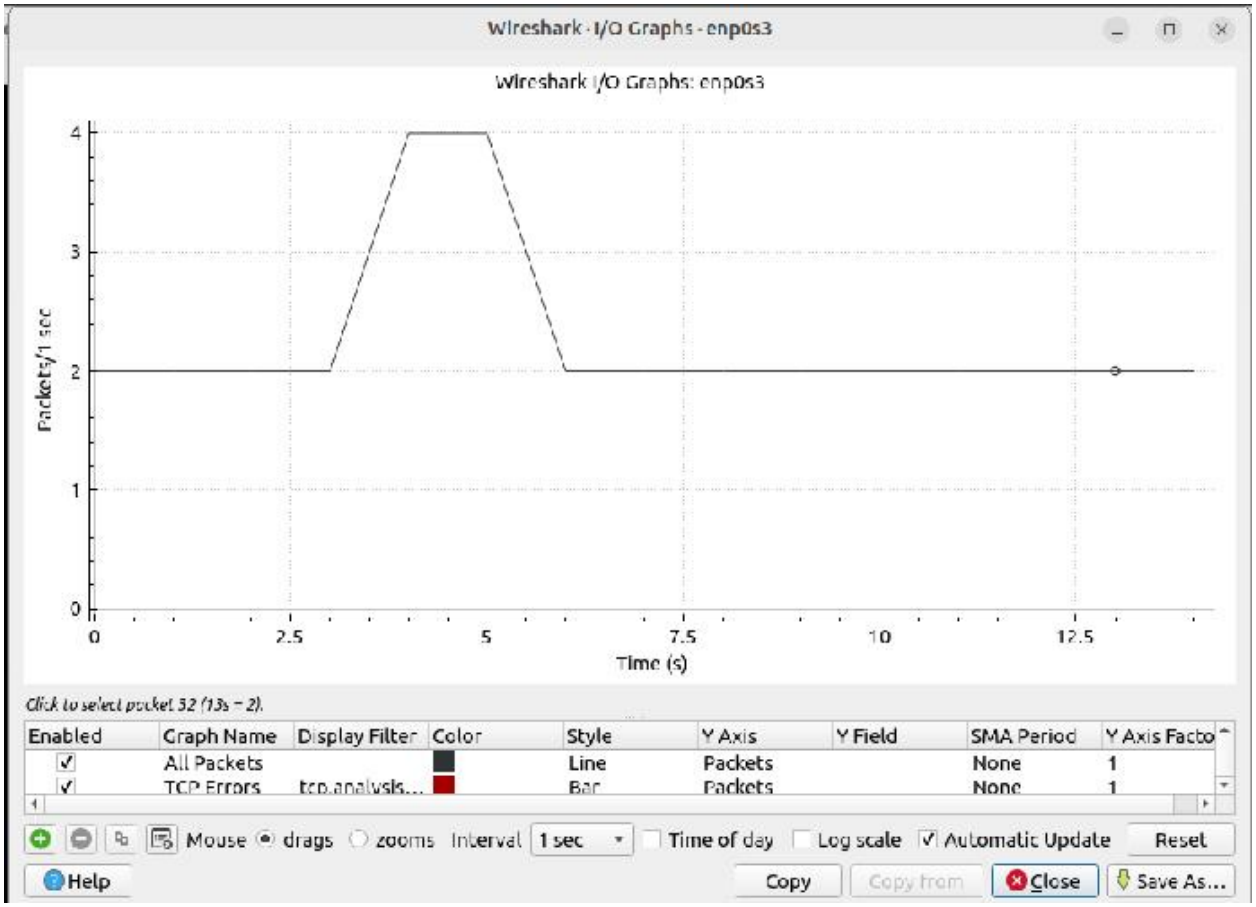
```
tcp.flags.syn==1 and tcp.flags.ack==0
```

This will show only SYN requests (which is characteristic of a SYN flood).

Summary:

Task	Command/Action
Install Snort	`sudo apt-get install snort`
Configure Snort	`sudo nano /etc/snort/snort.conf`
Start Snort IDS	`sudo snort -A console -c /etc/snort/snort.conf -i eth0`
Install hping3	`sudo apt-get install hping3`
Launch SYN Flood	`hping3 -S -p 80 192.168.56.101 --flood`
Monitor with Wireshark	Apply `tcp` or `tcp.flags.syn==1 and tcp.flags.ack==0` filter

NORMAL PINGING TO CHECK CONNECTIVITY



NOTE: ALWAYS USE OWASP VM FOR THIS hping3

RESULT OF `hping3 -S -p 80 192.168.56.101 -flood`

